

Module Interface Specification for Physiotherapy Companion

Team 11, technically functional

Matthew Huynh

Cieran Diebolt

Vaisnavi Shanthamoorthy

Maham Siddiqui

Eman Ashraf

April 14, 2026

1 Revision History

Date	Version	Notes
Nov 6 - Nov 14	1.0	Preliminary Draft
March 17 - ?	2.0	Starting final draft

2 Symbols, Abbreviations and Acronyms

See SRS Documentation at [SRS](#)

Contents

1	Revision History	i
2	Symbols, Abbreviations and Acronyms	i
3	Introduction	1
4	Notation	1
5	Module Decomposition	2
6	MIS	2
6.1	Module - Account Creation	2
6.1.1	Uses	3
6.1.2	Syntax	4
6.1.3	Semantics	6
6.1.4	Environment Variables	6
6.1.5	Assumptions	6
6.1.6	Access Routine Semantics - Exported	6
6.1.7	Local Functions	7
6.1.8	Access Routine Semantics - Local	8
6.2	Module - Valid user	8
6.2.1	Uses	9
6.2.2	Syntax	9
6.2.3	Semantics	10
6.2.4	Environment state variables	10
6.2.5	Assumptions	10
6.2.6	Access Routine Semantics - Exported	10
6.2.7	Local Routines	10
6.2.8	Access Routine Semantics - Local	11
6.3	Module - User Account	11
6.3.1	Uses	12
6.3.2	Syntax	12
6.3.3	Semantics	14
6.3.4	Environment Variables	15
6.3.5	Assumptions	15
6.3.6	Access Routine Semantics - Exported	15
6.3.7	Local Functions	16
6.3.8	Access Routine Semantics - Local	16
6.4	Module - Exercise Data	16
6.4.1	Uses	16
6.4.2	Syntax	20
6.4.3	Semantics	23

6.4.4	Environment Variables	23
6.4.5	Assumptions	24
6.4.6	Access Routine Semantics - Exported	24
6.4.7	Local Functions	25
6.4.8	Access Routine Semantics - Local	26
6.5	Module - User Activity	26
6.6	Module - Analysis Control Flow	26
6.6.1	Uses	26
6.6.2	Syntax	29
6.6.3	Semantics	31
6.6.4	Environment variables	32
6.6.5	Assumptions	32
6.6.6	Access Routine Semantics - Exported	32
6.7	Module - Analyze	34
6.7.1	Uses	34
6.7.2	Syntax	35
6.7.3	Semantics	36
6.8	Module - Feedback	37
6.9	Module - Metrics	37
6.10	Module - Draw	37
6.11	Module - UI	37
7	Appendix A	37

3 Introduction

The following document details the Module Interface Specifications for the PhysioCompanion app. The application is designed to identify users performing prescribed exercises and provide feedback on their form. This tool uses computer vision, specifically OpenCV and MediaPipe, to analyze user video recording and perform analysis. Please note that certain modules exhibit functional overlap, as their respective operations perform equivalent decomposed tasks across different modules.

Complementary documents include the [System Requirement Specifications](#) and [Module Guide](#). The full documentation and implementation can be found at [here](#).

4 Notation

The structure of the MIS for modules comes from [Hoffman and Strooper \(1995\)](#), with the addition that template modules have been adapted from [Ghezzi et al. \(2003\)](#). The mathematical notation comes from Chapter 3 of [Hoffman and Strooper \(1995\)](#). For instance, the symbol $:=$ is used for a multiple assignment statement and conditional rules follow the form $(c_1 \Rightarrow r_1 | c_2 \Rightarrow r_2 | \dots | c_n \Rightarrow r_n)$.

The following table summarizes the primitive data types used by Physiotherapy Companion.

Data Type	Notation	Description
character	char	a single symbol or digit
integer	$\mathbb{Z}$	a number without a fractional component in $(-\infty, \infty)$
natural number	$\mathbb{N}$	a number without a fractional component in $[1, \infty)$
real	$\mathbb{R}$	any number in $(-\infty, \infty)$
float	F	floating point numbers within (approx.) $(-1.798 \times 10^{308}, 1.798 \times 10^{308})$
object	X	an abstract representation of a collection of the above, similar to tuple
array	a	a collection of a single data type, n entries of the form $(a_0, a_1, a_2, \dots, a_{n-1})$
list	L	an array of non-fixed length

This specification of PhysioCompanion uses some derived data types: sequences, strings, and tuples. Sequences are lists filled with elements of the same data type. Strings are sequences of characters. Tuples contain a list of values, potentially of different types. In

addition, PhysioCompanion uses functions, which are defined by the data types of their inputs and outputs. Local functions are described by giving their type signature followed by their specification.

5 Module Decomposition

The following table is taken directly from the Module Guide document for this project.

Level 1	Level 2
Hardware-Hiding Module	
Behaviour-Hiding Module	User Interface Module Home Module Main Module Draw Module
Software Decision Module	Database Management Module Generator Module Metrics Module Analyze Module Feedback Module

Table 1: Module Hierarchy

6 MIS

This section describes individual module specifications.

6.1 Module - Account Creation

This module enables the creation of an account for users. It uses a database to store any information that is essential for other modules to perform their actions. A database was used for its ease of manipulation and its comprehensive, multi-functional nature.

The database tables related to this module are:

1. User data (**users**): This table consists of information such as name, role, email, birth-day, license_no/invite_code, createdAt and physioID.
2. Invite_codes (**inviteCodes**): This table contains all valid invite codes that will be used to validate patient account creation on the application.

3. Valid licenses (`validLicenses`): This is used for verification and validation purposes for a physiotherapist account creation.

6.1.1 Uses

/technically-functional/mobile/lib/authService.ts

{Imported Variables}

<code>auth</code>	Firestore database instance imported from <code>"/firebase"</code> . Used for all authentication operations including email checking, sign-in, sign-out, and account creation.
<code>db</code>	Firestore database instance imported from <code>"/firebase"</code> . Used for reading and writing user data, license validation, and invite code management.

{Imported Types}

<code>None</code>	No external types are imported in this file. All type definitions are inferred or defined inline.
-------------------	---

{Imported Functions}

`createUserWithEmailAndPassword`

Firestore Auth function from `"firebase/auth"`. Used to create new user accounts in Firestore Authentication for both physiotherapist and patient registration.

<code>doc</code>	Firestore function from <code>"firebase/firestore"</code> . Used to create document references for users, validLicenses, and inviteCodes collections.
------------------	---

<code>getDoc</code>	Firestore function from <code>"firebase/firestore"</code> . Used to fetch user documents, license documents, and invite code documents from Firestore.
---------------------	--

<code>setDoc</code>	Firestore function from <code>"firebase/firestore"</code> . Used to create new user documents in Firestore during registration.
---------------------	---

<code>updateDoc</code>	Firestore function from <code>"firebase/firestore"</code> . Used to mark invite codes as used after patient registration.
------------------------	---

`serverTimestamp`

Firestore function from `"firebase/firestore"`. Used to set createdAt timestamps on new user documents.

6.1.2 Syntax

Exported Constants

authService.ts

`export const registerPatient`

Async function constant that registers a new patient account. Validates invite code, checks email uniqueness, creates Firebase Auth user, and stores patient data in Firestore.

`export const registerPhysio`

Async function constant that registers a new physiotherapist account. Validates license number format and existence, checks email uniqueness, creates Firebase Auth user, and stores physio data in Firestore.

`export const checkEmailExists`

Async function constant that checks if an email address is already registered in the system by querying the Firestore users collection.

`export const licenseFormatValid`

Async function that validates the format of a physiotherapist license number against the pattern XX-123456.

Exported Access Programs

authService.ts

Input	Output	Exceptions
checkEmailExists(email): Checks if an email is already registered in Firebase Authentication.		
email: string - User's email address	Promise<boolean> - Returns true if email already exists, false otherwise	Error - Throws error if there is an error checking email.
licenseFormatValid(licenseNumber): Validates physiotherapist license number format (XX-123456).		
licenseNumber: string - PT license number	boolean - Returns true if format is valid, false otherwise	Error - Throws error if there is an error checking licenseNumber.
registerPhysio: Registers a new physiotherapist account into the database after license validation		

Continued on next page

Input	Output	Exceptions
params: {name: string, email: string, password: string, licenseNumber: string}	Promise<{uid: string, role: "physio"}> - Returns user ID and role	Error - Invalid license format, license not verified, Firebase Auth error
registerPatient: Registers a new patient account into the database.		
params: {name: string, email: string, password: string, inviteCode: string, birthday: string}	Promise<{uid: string, role: "patient"}> - Returns user ID and role, marks invite code as used	Error - Invalid invite code, inactive/used code, missing physio link, or Firebase Auth error

Table 2: Exported functions

Exported Access Programs

p: (name, email, password, licenseNumber, birthday)

pt: (name, email, password, inviteCode, birthday)

user: p or pt

e: user.email

n: user.name

pw: user.password

lno: user.licenseNumber

code: user.inviteCode

bday: user.birthday

cat: user.createdAt

pid: user.physioId (for pt only)

v: user.verified (for p only)

u: user.uid

del: user.deleted (for pt only)

db: **users** table in database

db': newer version of **users** table

$i, j, k, l, m, n \in \{0 - 9\}$

$a, b \in \{A - Z\}$

$E(e)$: 'checkEmailExists(e)'

$L(l)$: 'licenseFormatValid(l)'

$Phy(p)$: 'registerPhysio(p)'

$Pt(pt)$: 'registerPatient(pt)'

Function definitions:

$E(e) \implies e \in db.email$

$L(l) \implies \exists l \mid l = (a + b + '-' + 'i' + 'j' + 'k' + 'l' + 'm' + 'n') \in validLicenses$

$Phy(p) \implies db' = db \cup users \mapsto \{uid \mapsto \{cat, e, lno, n, role : 'physio', v\}\}$

$Pt(pt) \implies db' = db \cup users \mapsto \{uid \mapsto \{cat, del, e, code, n, pid, role : 'patient'\}\}$

6.1.3 Semantics

State Variables

See [here](#) for a list of additional state variables.

first: user 1

second: user 2

first_uid: uid of user 1

second_uid: uid of user 2

State Invariant

$\forall first, second \in db, first_uid \cap second_uid = \emptyset$

All users have different uids.

6.1.4 Environment Variables

screen: The application's display to the user

touchscreen: Users will click on the screen to select their desired action

6.1.5 Assumptions

- Patients with the same name have different birthdays.
- Only one person can access user information in the database at a time since many values that are exported are asynchronous values.

6.1.6 Access Routine Semantics - Exported

Exported access routines

See [here](#) for state variables being used.

checkEmailExists: $E(e)$

Pre-condition: $e \in db.email$

Function: $E(e) = true$

Post-condition: Returns true if email already exists in database(users), false otherwise.

licenseFormatValid: $L(l)$

Pre-condition: $l \in db.validLicenses$

Function: $L(l) = true$

Post-condition: Returns true if license exists in database(`validLicenses`), false otherwise.
registerPhysio: $Phy(p)$

Pre-condition: $(p.name \cap p.email \cap p.pw \cap p.lno \cap p.bday \neq \emptyset) \cap (L(p.lno)) \cap (\neg(E(p.email)))$

Function: $Phy(p)$

Post-condition: $p \in db$

registerPatient: $Pt(pt)$

Pre-condition: $(p.name \cap p.email \cap p.pw \cap p.code \cap p.bday \neq \emptyset) \cap (\neg(E(p.email)))$

Function: $Pt(pt)$

Post-condition: $p \in db$

6.1.7 Local Functions

Input	Output	Exceptions
passwordCheck(): Internal helper function for password validation.		
params: {password: string}	string null	• Returns "Password must be at least 6 characters long" if length < 6 • Returns "Password must contain at least one number" if no numbers present • Returns null if valid
validateBirthday(): Internal helper function for birthday format validation.		
params: {birthday: string}	string null	• Returns "Please enter birthday in YYYY-MM-DD format" if format entered is invalid • Returns null if valid

Table 3: Account module: Local validation functions

Note: CreatePatient and registerPatient have an important distinction although they seem similar. CreatePatient is the React component which manages the interface, while registerPatient handles the backend logic for making a new patient account. CreatePhysio and registerPhysio have the same distinction.

Local Routines (definitions)

p: (name, email, password, licenseNumber, birthday)

pt: (name, email, password, inviteCode, birthday)

user: p or pt

e: user.email

n: user.name

pw: user.password

lno: user.licenseNumber

code: user.inviteCode

bday: user.birthday

Pc(pw): 'passwordCheck(pw)'

Vb(bday): 'validateBirthday(bday)'

null: null value

Function definitions:

$Pc(pw) \implies pw.length \geq 6 \cap \exists d \mid d \in \{0 - 9\} \cap d \in pw$

$Date = 'YYYY-MM-DD' \mid 'YYYY' \in \{0000 - 9999\} \cap 'MM' \in \{01 - 12\} \cap 'DD' \in \{01 - 31\}$

$Vb(bday) \implies bday \in Date$

Note: See 6.11 for CreatePhysio and CreatePatient interface specifications.

6.1.8 Access Routine Semantics - Local

Vb(bday):

Pre-condition: $bday \notin \emptyset$

Function: $Vb(bday)$

Post-condition: returns null if valid

Pc(pw):

Pre-condition: $pw \notin \emptyset$

Function: $Pc(pw)$

Post-condition: returns null if valid

6.2 Module - Valid user

This module's purpose is to check whether a potential user is authorized to create an account. For feasible implementation purposes, a file with hard coded invite codes and license numbers will be used. The actions accomplishing the module's purpose are within the registerPhysio and registerPatient functions, which contain internal validation of inviteCode and licenseNumber.

6.2.1 Uses

<code>params</code>	Input object containing patient registration data. Type: <code>{name: string, email: string, password: string, inviteCode: string, birthday: string}</code> .
<code>code</code>	Normalized invite code. Type: <code>string</code> . Computed as <code>params.inviteCode.trim().toUpperCase()</code> .
<code>inviteRef</code>	Firestore document reference to <code>inviteCodes/code</code> . Type: <code>DocumentReference</code> .
<code>inviteSnap</code>	Firestore document snapshot. Type: <code>DocumentSnapshot</code> .
<code>invite</code>	Data from invite code document. Contains fields: <code>active</code> (boolean), <code>used</code> (boolean), <code>physioId</code> (string).
<code>physioId</code>	uid of the physiotherapist that provided invite code. Type: <code>string</code> .
<code>cred</code>	Firebase Auth user credential. Type: <code>UserCredential</code> .
<code>uid</code>	Firebase Authentication unique user ID. Type: <code>string</code> .
<code>db</code>	Firestore database instance. Imported from <code>./firebase</code> .
<code>auth</code>	Firebase Authentication instance. Imported from <code>./firebase</code> .
<code>licenseNumber</code>	State variable managing license number input. Type: <code>string</code> . Initialized as empty string.
<code>lic</code>	Normalized license number. Type: <code>string</code> . Computed as <code>params.licenseNumber.trim().toUpperCase()</code> .
<code>licSnap</code>	Firestore document snapshot from <code>validLicenses</code> collection. Type: <code>DocumentSnapshot</code> .

6.2.2 Syntax

Exported Constants

See ***authService.ts*** exported constants in `??`.

For interface elements, see [6.11](#).

Exported Access Programs

See [Table 2](#) for ***registerPatient*** and ***registerPhysio*** details respectively.

Function definitions:

This module has components that have been defined in the Account Creation module. See [function definitions in 6.1.2](#).

6.2.3 Semantics

State Variables

See [6.11](#) for UI state variables.

pt: a patient user type

p: a physio user type

code: a patient invite code

lno: a PT's license number

users: `users` table in database

State Invariant

$code['used'] = true \implies \exists pt \mid pt.inviteCode == code$

$\exists p.uid \in users \implies p.licenseNumber \in validLicenses$

6.2.4 Environment state variables

Not applicable

6.2.5 Assumptions

- All practicing physiotherapists have an active and valid license.
- All active invite codes are used by new and active patients only.
- Invite codes and license numbers are unique.

6.2.6 Access Routine Semantics - Exported

Exported access routines

See [account creation semantics](#) for details of respective functions.

6.2.7 Local Routines

Portions that are pertinent to the module are elaborated on below.

registerPatient: Internal Validation Steps

The function performs the following sequential validations on the invite code:

1. **Existence Check:** Query Firestore `inviteCodes` collection for document with ID = `normalize(inviteCode)`. If not found, throw "Invalid invite code."
2. **Active Check:** If `invite.active` $\neq$ `true`, throw "Invite code inactive."
3. **Usage Check:** If `invite.used` = `true`, throw "Invite code already used."
4. **Physio Link Check:** If `physioId` = `null` or does not exist, throw "Invite code missing physio link."

Only after all validations pass does the function proceed to:

1. Create Firebase Auth user
2. Create Firestore user document with `role:"patient"` and `uid`
3. Mark invite code as used with `usedBy:uid`

registerPhysio: Internal Validation Steps

The function performs the following sequential validations on the license number:

1. **Format Check:** Validate license number format using `licenseFormatValid()`. License must match pattern: two uppercase letters, a dash, and six digits (e.g., ON-123456). If format is invalid, throw "Invalid license format (example: ON-123456)."
2. **Existence Check:** Query Firestore `validLicenses` collection for document with ID equal to the normalized license number. If not found, throw "License not verified."
3. **Active Check:** If the license document has `active` field not equal to true, throw "License not verified."

Only after all validations pass does the function proceed to:

1. Create Firebase Auth user using `createUserWithEmailAndPassword()`
2. Create Firestore user document with `role:"physio"`, `verified:true`, and `uid`
3. Return `{uid, role:"physio"}`

6.2.8 Access Routine Semantics - Local

NA [Account creation access routine semantics](#)

6.3 Module - User Account

This module's purpose is facilitate data manipulation within the user database tables. It was created as a separate component to simplify implementation. Account creation generates a single instance of one of the following objects:

- **Patient object:** `{uid, email, name, role: "patient", birthday?, physioId?, inviteCode?, createdAt?, deleted?}`
- **Physio object:** `{uid, email, name, role: "physio", licenseNumber?, verified?, createdAt?}`

6.3.1 Uses

useraccount.ts

uid: `string`; Unique identifier for the user. Used for fetching user data, updating user information, deleting user accounts, and associating patients with physiotherapists.

email: `string`; User's email address for authentication and communication. Used for email existence verification, credential validation, and user authentication.

name: `string`; User's full name. Used for searching users by name, retrieving user database information by name, and patient account deletion verification.

role: `"patient" | "physio"`; User's role in the system. Used for filtering patients by physiotherapist and categorizing users during database information retrieval.

physioId?: `string`; Reference to the physiotherapist assigned to a patient. Used for retrieving all patients belonging to a specific physiotherapist.

deleted?: `boolean`; Soft delete flag indicating whether the user account has been deleted. Used as a filter when querying active patients and during user account deletion operations.

birthday?: `string`; User's birthday. Used as part of patient profile information.

licenseNumber?: `string`; Professional license number for physiotherapists. Used as part of physiotherapist profile information.

verified?: `boolean`; Flag indicating whether a physiotherapist's license has been verified. Used in physiotherapist profile information.

inviteCode?: `string`; Invitation code for patient registration. Used as profile information and account creation validation.

createdAt?: `any`; Timestamp of user account creation. Defined in the interface but not set or read by current methods. Recommended for future audit trail implementation.

6.3.2 Syntax

Exported Constants

None

Exported Access Programs

useraccount.ts

Input	Output	Exceptions
validateCredentials(email, password): Authenticates user with Firebase Auth.		
email: string, password: string	Promise<boolean>	None - returns false on failure
getUserData(uid): Retrieves user data from Firestore by UID.		
uid: string	Promise<UserData null>	UserAccountFieldEmptyError
getUserByEmail(email): Retrieves user data from Firestore by email.		
email: string	Promise<UserData null>	None
emailExists(email): Checks if an email is already registered.		
email: string	Promise<boolean>	None
updateUser(uid, updates): Updates user data in Firestore.		
uid: string, updates: Partial<UserData>	Promise<boolean>	UserAccountNotFound
deleteUser(uid): Soft-deletes a user by setting deleted flag to true.		
uid: string	Promise<boolean>	None
deleteUserByEmail(email): Soft-deletes a user by email address.		
email: string	Promise<boolean>	UserAccountNotFound
getPatientsByPhysio(physioId): Retrieves all active patients assigned to a physiotherapist.		
physioId: string	Promise<UserData[]>	None
getUsersByName(name): Searches users by partial name match.		
name: string	Promise<UserData[]>	None
getAllUsers(): Retrieves all users from Firestore.		
None	Promise<UserData[]>	None
getUserdbInfo(name, birthday?): Retrieves users by name and optional birthday, separated by role.		
name: string, birthday?: string	Promise<{patients: UserData[], physios: UserData[]}>	UserAccountNotFound
PTaccountDelete(physioEmail, patientName, patientEmail): Allows physiotherapist to soft-delete a linked patient account.		

Continued on next page

Input	Output	Exceptions
physioEmail: string, patientName: string, patientEmail: string	Promise<void>	UserAccountField EmptyError UserAccountNot FoundError
usernamePwMatch(email, password): Verifies email and password combination.		
email: string, password: string	Promise<boolean>	UserAccountLogin MatchError
authenticateUser(email, password): Authenticates user and returns user data if successful.		
email: string, password: string	Promise<UserData null>	None
initializeSystem(): Initializes the system (placeholder for future setup).		
None	Promise<void>	None

Table 4: UserAccount module: Exported access programs

6.3.3 Semantics

State Variables

useraccount.ts

private usersCollection: string Internal state variable storing the Firestore collection name for user documents.

private validLicensesCollection: string Internal state variable storing the Firestore collection name for physiotherapist licenses.

private inviteCodesCollection: string Internal state variable storing the Firestore collection name for invitation codes.

State Invariant

users: 'users' database table

u1: user 1

u2: user 2

uid: uid

email: email

Phy: set of all users that have role = 'physio'

Pt: set of all users that have role = 'patient'

U : set of all UserData objects
 E : set of valid emails **updates**: part of UserData object
 $\forall u1, u2 \in users, u1 \neq u2 \implies u1.email \neq u2.email$
 $\forall u1 \in Phy, u1.physioId = \emptyset$
 $\forall u1 \in Phy, u1.verified = true$
 $\forall u1 \in Pt, u1.deleted = true \implies u1.uid \in \emptyset$
 $\forall u1 \in P, u2 \in Phy, \exists u1.physioId = u2.uid$
 $\forall id \in \{users.uid\}, \exists u1 | u1.uid = id \cap u1 \in users$
 $\forall u1 \in users \cap u1 \in Phy \implies u1 \notin P$
 $\forall u2 \in users \cap u2 \in Pt \implies u2 \notin Phy$

6.3.4 Environment Variables

Not applicable

6.3.5 Assumptions

- If two people have the same name, then it is assumed they have different different roles.
- No two PTs have the same license numbers.

6.3.6 Access Routine Semantics - Exported

Note: Auth functions access routine semantics are not outlined here.

getUserData(uid)

Pre-condition: $uid \notin \emptyset$

Function: getUserData(uid)

Post-condition: returns UserData object — $uid = user.uid$

getUserByEmail(email)

Pre-condition: $email \notin \emptyset$

Function: getUserByEmail(email)

Post-condition: returns UserData object — $email = user.email$

emailExists(email)

Pre-condition: email is a valid email

Function: emailExists(email)

Post-condition: $True \implies email \in E$

updateUser(uid, updates)

Pre-condition: $user \in users | \exists user.uid = uid$

Function: emailExists(email)

Post-condition: $True \implies email \in E$

deleteUser(uid)

Pre-condition: $user \in users$

Function: `deleteUser(uid)`

Post-condition: $users' = users - (user[uid])$

deleteUserByEmail(email)

Pre-condition: $user \in users$

Function: `deleteUserByEmail(email)`

Post-condition: $users' = users - (user[uid])$

6.3.7 Local Functions

Not applicable. All helper functions are defined as private class functions.

6.3.8 Access Routine Semantics - Local

Not applicable

6.4 Module - Exercise Data

This module manages exercises that will be performed by the patient. It consists of a data layer (*exerciseData.ts*) and a UI layer (*exercises.tsx*, *[id].tsx*). The data layer provides database operations using the `userExercises` table. The UI layer fulfills the purpose of interfacing between the user and the module and is discussed in 6.11. This structure was chosen to enable:

- High cohesion: all constituents serve the same business purpose
- Low coupling: there is no direct communication present between the front end and database. They communicate through the backend.
- Easy maintenance: low coupling allows for maintenance of a single 'part' of the module without affecting the rest of the module.

6.4.1 Uses

exerciseData.ts - Functions

`collection` Firestore function from "firebase/firestore". Used to reference a database collection for exercise operations.

`deleteDoc` Firestore function from "firebase/firestore". Used to delete an exercise document from the `userExercises` collection.

doc Firestore function from "firebase/firestore". Used to create a document reference when fetching a single exercise.

getDoc Firestore function from "firebase/firestore". Used to fetch a single exercise document by ID.

getDocs Firestore function from "firebase/firestore". Used to fetch multiple exercise documents.

query Firestore function from "firebase/firestore". Used to create filtered queries on exercise collections.

where Firestore function from "firebase/firestore". Used to specify filter conditions (userid, title) in exercise queries.

addDoc Firestore function from "firebase/firestore". Used to add a new exercise document to the userExercises collection.

updateDoc Firestore function from "firebase/firestore". Used to update exercise fields (description, reps, sets, title) in an existing document.

exerciseData.ts - Variables

db Firestore database instance from ".firebase". Used as the database reference for all Firestore operations related to this file.

exerciseData.ts - Types

Exercise (type) Type definition defined locally within this file. Represents the exercise data structure with properties: id, title, subtitle, description, demoText, enabled, tags, sets, reps.

exercises.tsx - Functions

useEffect React hook from "react". Used to perform side effects when user data, role, or selected user changes.

useState React hook from "react". Used to manage local state for user data, exercises, modal visibility, and form values.

getExercisesById Function from *exerciseData.ts*. Used to fetch patient's assigned exercises for the patient view.

getGeneralExercises Function from *exerciseData.ts*. Used to fetch the master exercise library for physiotherapists.

getSelectedExercises Function from *exerciseData.ts*. Used to fetch currently assigned exercises for checkbox state and modal preloading.

addUserExercise Function from *exerciseData.ts*. Used to assign an exercise to a patient when a physiotherapist checks a checkbox.

removeUserExercise Function from *exerciseData.ts*. Used to remove an exercise assignment when a physiotherapist unchecks a checkbox.

updateUserExercise Function from *exerciseData.ts*. Used to save modified exercise instructions (description, sets, reps).

getCurrentUser Function from "@lib/temp". Used to fetch the currently authenticated user on component mount.

getUserRole Function from "@lib/roleStore". Used to determine if the current user is a patient or physiotherapist.

getSelectedUserID Function from "@lib/profileActivity". Used to get the patient ID selected by a physiotherapist.

exercises.tsx - Variables

userData State variable storing current authenticated user. Initialized via `getCurrentUser()`.

userRole State variable storing user role. Initialized via `getUserRole()`.

selectedUser
State variable storing selected patient ID for physio view. Initialized via `getSelectedUserID()`.

userExercises
State variable storing patient's assigned exercises. Initialized via `getExercisesById()`.

selectedIds
State variable storing titles of checked exercises. Initialized via `getSelectedExercises()`.

physioExercises
State variable storing master exercise library. Initialized via `getGeneralExercises()`.

roleReady Boolean state variable indicating role has been loaded.

modalVisible
Boolean state variable controlling modal visibility.

modalText State variable storing custom description text in modal.

modalSets State variable storing custom sets value in modal.

modalReps State variable storing custom reps value in modal.

modalExercise
 State variable storing exercise being modified in modal.

exercises.tsx - Types

Exercise Type from "../lib/exerciseData". Used to type exercise objects, exercise lists, and modal exercise state.

UserData Type from "@lib/useraccount". Used to type the current user data state.

UserRole Type from "@lib/roleStore". Used to type the user role state ("patient" — "physio").

[id].tsx - UI Functions (React Native & Navigation)

useLocalSearchParams
 Expo Router hook from "expo-router". Used to extract the id parameter from the navigation route.

useRouter Expo Router hook from "expo-router". Used for navigation (goBack and navigate to record screen).

useVideoPlayer
 Expo Video hook from "expo-video". Used to create and configure the video player instance for exercise demonstration videos.

VideoView Component from "expo-video". Used to render the video player UI for exercise demonstrations.

View React Native component from "react-native". Used as a container for layout.

Text React Native component from "react-native". Used to display exercise title, subtitle, description, sets, reps, and UI text.

StyleSheet React Native function from "react-native". Used to create component styles.

Pressable React Native component from "react-native". Used to create touchable buttons (Record, Back, Go Back).

ScrollView React Native component from "react-native". Used to enable scrolling for exercise details content.

require React Native asset loading function. Used to load the local demonstration video file from previously stored tutorial video.

[id].tsx - React Hooks & Data(Functions)

useEffect React hook from "react". Used to fetch exercise data when the component mounts or when the id parameter changes.

useState React hook from "react". Used to manage the currentExercise state variable.

getExercise
Function from "../..lib/exerciseData". Used to fetch a single exercise document by its ID from the Firestore database. This is a data access function, not UI-related.

[id.tsx] - Variables

id Route parameter extracted from useLocalSearchParams. Contains the exercise ID passed from navigation. Used to fetch the specific exercise from the database.

videoSource
Local variable that conditionally loads a video asset. Returns require() path only for "Seated Knee Flexion-Extension Exercise", otherwise returns null.

player Video player instance created by useVideoPlayer. Used to control video playback (loop, play, pause) in the VideoView component.

[id.tsx] - Imported Types

Exercise Type definition imported from "../..lib/exerciseData". Represents the exercise data structure with properties: id, title, subtitle, description, demoText, enabled, tags, sets, reps. Used to type the currentExercise state.

6.4.2 Syntax

Exported Constants

No exported constants are present in any of the reviewed files (exerciseData.ts, exercises.tsx, [id].tsx.).

Exported Access Programs

exerciseData.ts

Input	Output	Exceptions
getExercisesById(uid): Retrieves all exercises assigned to a specific user from the "userExercises" collection.		
uid: string - User ID of the patient	Promise<Exercise[] undefined> - Returns array of exercises or undefined if none found	Error - Throws error if there is an issue fetching exercises from Firestore
getExercise(exid): Fetches a single exercise document by its ID.		
exid: string - Exercise document ID	Promise<Exercise undefined> - Returns exercise object or undefined if not found	Error - Throws error if there is an issue fetching the exercise from Firestore
getGeneralExercises(): Retrieves the master exercise library from the "exercises" collection.		
None	Promise<Exercise[] undefined> - Returns array of all general exercises or undefined if none found	Error - Throws error if there is an issue fetching exercises from Firestore
getSelectedExercises(uid): Fetches the current list of exercises assigned to a specific user.		
uid: string - User ID of the patient	Promise<Exercise[] undefined> - Returns array of assigned exercises or undefined if none	Error - Throws error if there is an issue fetching exercises from Firestore
removeUserExercise(uid, exerciseTitle): Deletes an exercise assignment from a user's exercise list.		
uid: string - User ID of the patient, exerciseTitle: string - Title of the exercise to remove	Promise<void> - Returns nothing on successful deletion	Error - Throws error if there is an issue deleting the document from Firestore
addUserExercise(uid, exercise): Creates a new exercise assignment for a specific user.		
uid: string - User ID of the patient, exercise: Exercise - Exercise object to assign	Promise<void> - Returns nothing on successful creation	Error - Throws error if there is an issue adding the document to Firestore

Continued on next page

Input	Output	Exceptions
updateUserExercise(uid, exerciseTitle, exerciseDescription, exerciseSets, exerciseReps): Modifies an existing user exercise's instructions (description, sets, reps).		
uid: string - User ID of the patient, exerciseTitle: string - Title of the exercise, exerciseDescription: string - New description, exerciseSets: string - Number of sets (optional), exerciseReps: string - Number of reps (optional)	Promise<void> - Returns nothing on successful update	Error - Throws error if there is an issue updating the document in Firestore

Table 5: Exported Access Programs - exerciseData.ts

exercises.tsx

Input	Output	Exceptions
ExercisesTab(): Default exported React component that renders the exercises interface for both patients and physiotherapists.		
None - Takes no props	JSX.Element - Returns a React component with role-based views (patient exercise list or physiotherapist exercise selection interface)	None - No exceptions thrown; returns null while role is loading

Table 6: Exported Access Programs - exercises.tsx

[id].tsx

There are no exported access programs in this file. The file exports only the default React component `ExerciseDetail`, which is a presentation component rather than an access program.

The file **consumes** the following access programs from `exerciseData.ts`:

- `getExercise()` - Fetches exercise data by ID

6.4.3 Semantics

State Variables

userexercises: userExercises table in the database

exercises: exercises table in user database

users: users table in the database

ue: modified exercise object

uue: unmodified exercise object

assigned: list of assigned exercises

u1: user

exid: exercise id

ex: exercise (object)

State Invariant

$\forall e1, e2 \in exercises, e1 \neq e2 \implies e1.title \neq e2.title \cap e1.description \neq e2.description$

$\forall e1 \in exercises, \exists e1.reps, e1.sets \in \mathbb{Z}^+$

$\forall e1, e2 \in userexercises, e1.title = e2.title \not\Rightarrow (e1.id = e2.id \cup e1.reps = e2.reps \cup e1.sets = e2.sets \cup e1.userid = e2.userid)$

$\forall ue, e1 \in userexercises \mid ue.userid = e1.userid \cap ue.title = e1.title, \exists ue.description \neq e1.description \cup ue.sets \neq e1.sets \cup ue.reps \neq e1.reps$

$\forall user \in users, |assigned| \geq 0$

Function definitions

u1: user

uid: uid (e.g. u1.uid)

title: exercise title

Ux: 'getExercisesById(uid)'

Ge: 'getExercise(exid)'

Gen: 'getGeneralExercises()'

Sel: 'getSelectedExercises(uid)'

R: 'removeUserExercise(uid, ex.title)'

A: 'addUserExercise(uid, ex)' $Ux(uid): \forall ex \in userexercises Ux(uid) = [ex] \mid ex.userid = uid$

$Ge(exid): \forall ex \in exercises, Ge(exid) = [ex] \mid ex.id = exid$

$Gen(): Gen() \rightarrow \{ex\} \in exercises$

$Sel(uid): \forall ex \in userexercises, Sel(uid) = [ex] \mid ex.userid = uid$

$R(uid, title): userexercises' = userexercises - [ex] \mid ex.title = title \cap ex.userid = uid$

$A(uid, ex): userexercises' = userexercises \cup [ex] \mid ex \in exercises \cap (ex \in userexercises \cap ex.userid = uid)$

6.4.4 Environment Variables

Not applicable

6.4.5 Assumptions

- For the same user, no two exercises with the same name are present. Conversely, no two exercises with the same names can be assigned by a PT to one patient.

6.4.6 Access Routine Semantics - Exported

db: database

userexercises: `userExercises` table in the database

exercises: `exercises` table in user database

users: `users` table in the database

ue: modified `exercise` object

assigned: list of assigned exercises

u1: user

exid: exercise id

ex: exercise (object)

getExercisesById

Pre-condition: $u1.uid = ex.userid \in \{userexercises.userid\}$

Function: `getExercisesById(u1.uid)`

Post-condition: $[ex]$

getExercise

Pre-condition: $exid \in \{exercises.id\} \cap exid \neq \emptyset$

Function: `getExercises(ex.exid)`

Post-condition: $ex \iff ex.id = exid$

getGeneralExercises

Pre-condition: $\exists exercises \in db$ **Function:** `getGeneralExercises()`

Post-condition: $[ex] \neq \emptyset$

getSelectedExercises

Pre-condition: $u1 \in users \cap u1.uid = ex.userid$

Function: `getSelectedExercises(u1.uid)`

Post-condition: $[ex]$

removeUserExercise

Pre-condition: $u1 \in users \cap u1.uid = ex.userid \cap title = ex.title \in userexercises$ **Function:** `getSelectedExercises(u1.uid, ex.title)`

Post-condition: $user[exercises]' = user[exercises] - ex$

addUserExercise

Pre-condition: $u1 \in users \cap ex \in exercises$

Function: addUserExercise(u1.uid, ex)

Post-condition: $user[exercises]' = user[exercises] + ex$

updateUserExercise

Pre-condition: $u1 \in users \cap ex \in exercises \cap ex.userid = u1.uid$

Function: addUserExercise(u1.uid, ex.title, ex.description, ex.sets, ex.reps)

Post-condition: $ue = ex \cap (ex(u1.uid, ue.title) \cup ex(u1.uid, ue.description) \cup ex(u1.uid, ue.sets) \cup ex(u1.uid, ue.reps))$

6.4.7 Local Functions

exerciseData.ts

There are no local routines/helper functions present in this file.

exercises.tsx

These functions are UI related. Note that they are included here for readability. They are also in section 6.11.

Input	Output	Exceptions
openExercise(ex): Navigates to the exercise detail page for the selected exercise.		
ex: Exercise - The exercise object to view	void - Navigates to <code>"/exercise/[id]"</code> route using <code>router.push()</code>	None
toggleSelection(ex): Toggles exercise selection for a patient (physiotherapist view only).		
ex: Exercise - The exercise to toggle	void - Updates <code>selectedIds</code> state and calls <code>onChecked()</code> or <code>onUnchecked()</code>	None
onChecked(ex): Assigns an exercise to the selected patient.		
ex: Exercise - The exercise to assign	void - Calls <code>addUserExercise(selectedUser, ex)</code> to save to database and logs selection	None
onUnchecked(id): Removes an exercise assignment from the selected patient.		

Continued on next page

Input	Output	Exceptions
id: <code>string</code> - The exercise title to remove	void - Calls <code>removeUserExercise(selectedUser, id)</code> to delete from database and logs deselection	None
modifyInstructions(ex): Opens modal to edit exercise instructions for a patient.		
ex: <code>Exercise</code> - The exercise to modify	Promise<void> - Sets <code>modalExercise</code> state, fetches existing instructions via <code>getSelectedExercises()</code> , and opens modal with preloaded values	None
updateInstructions(ex, newDesc, newSets, newReps): Saves modified exercise instructions to the database.		
ex: <code>Exercise</code> - The exercise being modified, newDesc: <code>string</code> - New description, newSets: <code>string</code> - New sets value, newReps: <code>string</code> - New reps value	void - Calls <code>updateUserExercise(selectedUser, ex.title, newDesc, newSets, newReps)</code> and logs the update	None

Table 7: Local Routines - exercises.tsx

6.4.8 Access Routine Semantics - Local

Not Applicable: they are UI related.

6.5 Module - User Activity

6.6 Module - Analysis Control Flow

This module is responsible for transmitting frames for analysis and the reverse.

6.6.1 Uses

server.py

server.py - Functions

Flask	From "flask". Used to create the Flask web application instance and define API routes.
jsonify	From "flask". Used to convert Python dictionaries to JSON responses for API endpoints.
request	From "flask". Used to parse incoming HTTP request data, including JSON payloads and form data.
CORS	From "flask_cors". Used to enable Cross-Origin Resource Sharing, allowing the mobile app to communicate with the backend.
threading.Lock	From Python standard library "threading". Used to create a lock for thread-safe access to the pose backend state.
base64.b64decode	From Python standard library "base64". Used to decode base64-encoded image strings sent from the mobile app.
np.frombuffer	From "numpy". Used to convert decoded image bytes into a NumPy array for image processing.
cv.imdecode	From "cv2" (OpenCV). Used to decode a NumPy array into an OpenCV image format (BGR).
cv.cvtColor	From "cv2" (OpenCV). Used to convert images between color spaces (BGR to GRAY) for brightness calculation.
cv.resize	From "cv2" (OpenCV). Used to resize frames to a standard dimension (640x480) before processing.
cv.flip	From "cv2" (OpenCV). Used to mirror images horizontally when using the front camera.
os.path.dirname	From Python standard library "os". Used to get the directory path of the current file.
os.path.abspath	From Python standard library "os". Used to get the absolute path of the current file.

`os.path.join` From Python standard library "os". Used to construct the full file path to the pose landmarker model.

`os.environ` From Python standard library "os". Used to set environment variables (OPENCV_VIDEOIO).

`exists` From "os.path". Used to check if the pose landmarker model file exists at the specified path.

`RuntimeError` Python built-in exception. Used to raise an error when the model file is missing.

`threading` Python standard library module. Used for thread management and locks.

server.py - Variables

`app` Flask application instance created for running the server.

`pose_app` Global instance of the PoseBackend class that manages pose detection state.

`here` Variable storing the directory path of the current Python file.

`model_path` Variable storing the full file system path to the pose_landmarker_heavy.task model file.

server.py - Types

`None` No external types are explicitly imported. Type hints use built-in types (str, bool, float, int) and OpenCV/NumPy types implicitly.

poseService.ts

poseService.ts - Functions

`axios.post` From "axios". Used to send HTTP POST requests to the backend server for frame processing, pre-checking, and resetting the backend state.

poseService.ts - Variables

`SERVER_URL` Constant string variable storing the backend server address and port (http://10.0.0.34 : 5001). Used as the base URL for all API calls to the Flask server.

`mirrored` Local boolean variable within processFrame and precheckFrame functions. Set to true when facing is "front", indicating the image needs to be mirrored.

<code>res</code>	Local variable storing the axios response object containing the backend response data.
<code>headers</code>	Configuration object passed to <code>axios.post</code> specifying Content-Type as "application/json".
<code>timeout</code>	Configuration value set to 15000ms for <code>processFrame</code> and <code>resetBackend</code> , 10000ms for <code>precheckFrame</code> .

poseService.ts - Types

<code>Side</code>	Type union defining valid knee sides: "RIGHT" — "LEFT". Used to specify which knee is being analyzed.
<code>Facing</code>	Type union defining camera orientation: "front" — "back". Used to determine if image mirroring is needed.
<code>Landmark</code>	Type defining a 3D pose landmark with x, y, z coordinates (all numbers). Represents a detected body point.
<code>Connection</code>	Type defining a tuple [number, number] representing an edge between two landmark indices. Used to draw skeleton connections.

`ProcessFrameResponse`

Type defining the response structure from the `process_frame` endpoint, containing landmarks, connections, metrics, and optional error.

`PrecheckFrameResponse`

Type defining the response structure from the `precheck_frame` endpoint, containing ok status, lighting and visibility flags, and message.

6.6.2 Syntax

Exported Constants

poseService.ts

There are no exported constants.

server.py

NA

Exported Access Programs

server.py

Input	Output	Exceptions
reset_backend(): Flask route handler for POST /reset endpoint. Resets pose analyzer state and restarts calibration.		
None	JSON - Returns {ok: true}	Exception - Throws exception if reset fails
precheck_frame(): Flask route handler for POST /precheck_frame endpoint. Validates lighting, frame composition, and knee visibility.		
JSON body: imageBase64: string, side: string, mirrored: boolean, legsOnly: boolean	JSON - Returns {ok, tooDark, inFrame, kneeVisible, message} or error response	Exception - Throws exception if frame decoding fails or processing error occurs
process_frame(): Flask route handler for POST /process_frame endpoint. Processes frame for pose detection and metric calculation.		
JSON body: imageBase64: string, side: string, mirrored: boolean, legsOnly: boolean	JSON - Returns {landmarks, connections, metrics} or error response with default metrics	Exception - Throws exception if frame decoding fails or processing error occurs

Table 8: Exported Access Programs - server.py

poseService.ts

Input	Output	Exceptions
processFrame(imageBase64, side, facing): Sends a single video frame to the backend for pose detection and metric calculation.		
imageBase64: string - Base64 encoded image frame, side: Side - "RIGHT" or "LEFT", facing: Facing - "front" or "back"	Promise<ProcessFrame> - Returns landmarks, connections, and metrics	Error - Throws error if request fails or timeout occurs (15000ms)
precheckFrame(imageBase64, side, facing): Performs pre-recording environment check including lighting and knee visibility.		

Continued on next page

Input	Output	Exceptions
imageBase64: string - Base64 encoded image frame, side: Side - "RIGHT" or "LEFT", facing: Facing - "front" or "back"	Promise<PrecheckFrame> - Returns ok status, lighting flags, visibility flags, and message	Error - Throws error if request fails or timeout occurs (10000ms)
resetBackend(): Resets the backend pose analyzer state and restarts calibration.		
None	Promise<void> - Resolves when reset is complete	Error - Throws error if request fails or timeout occurs (15000ms)

Table 9: Exported Access Programs - poseService.ts

6.6.3 Semantics

State Variables

F : set of video frames
 C : calibrating
 cal_time_left : time left for calibration
 P : frame responses set
 L : set of landmark arrays
 url : set of backend urls
 $T_{process}$: processFrame timeout value = 15000 ms
 $T_{precheck}$: precheckFrame timeout value = 10000 ms
 T_{reset} : timeout value for resetBackend() = 15000 ms

State Invariant

$\forall b \in \text{Base64} : b64_to_bgr_image(b) \in F$
 $\forall$ operations on pose_app : lock must be acquired first
 $resetBackend() \implies analyzer = newAnalyzer() \wedge c = true \cap cal_time_left = 10.0$
 $SERVER_URL \in url \cap SERVER_URL$ is immutable at runtime
 $\forall response \in P : response.landmarks \in l \cup \{\emptyset\} \cap response.connections \in C$
 $true \implies T_{process} = 15000 \cap T_{precheck} = 10000 \cap T_{reset} = 15000$
 $processFrameendpoint = SERVER_URL + /process_frame$
 $precheckFrameendpoint = SERVER_URL + /precheck_frame$
 $resetBackendendpoint = SERVER_URL + /reset$

6.6.4 Environment variables

NA

6.6.5 Assumptions

- All frames resize to 640x480 pixels.
- When `mirrored = true` (front camera), the frame is horizontally flipped. This assumes the patient is facing the camera and wants natural left-right orientation.
- A single user's frames can be processed at a time (this falls under the overarching assumption that the application can be used by one person at a time only).
- Images are assumed to be base64-encoded JPEG or PNG format. No other formats are supported.
- Timeout values assume reasonable response times.
- Reset function is assumed to be called before re-recording or new recording.

6.6.6 Access Routine Semantics - Exported

`f`: set of video frames
`c`: calibrating
`cal_time_left`: time left for calibration
`p`: frame responses set
`l`: set of landmark arrays
`url`: set of backend urls
`b`: set of base64 encoded frames
`s`: 'RIGHT' — 'LEFT'
`side`: $\in s$ facing: 'front' — 'back'
`response`: PrecheckFrameResponse
`J`: set of all JSON strings
`pose_app`: PoseBackend()

poseService.ts

processFrame:

Pre-condition: $\forall j \in J : j.imageBase64 \in b \cap j.side \in s \cap j.mirrored \in \{true, false\} \cap cameraReady = true \cap pose_app.analyzer \neq \emptyset$

Function: `processFrame(fr, side, facing)`

Post-condition: $fr \in f \implies pose_app.analyzer \neq \emptyset \cap \{rep_count\}(t_2) \geq \{rep_count\}(t_1) \forall t_2 > t_1, rom_degree = max_degree - min_degree \geq 0 \cap angle \in [0, 180] \cap c(t_2) = false \implies$

$c(t_1) = false \cup cal_time_left(t_1) \leq 0$ *precheckFrame*:

Pre-condition: $\forall j \in J : j.imageBase64 \in b \cap j.side \in s \cap j.mirrored \in true, false \cap cameraReady = true \cap pose_app.analyzer \neq \emptyset$

Function: *precheckFrame*(fr, s, facing)

Post-condition: $\forall j \in J : response.ok \in \{true, false\} \cap response.message \neq \emptyset : (response.ok = true \implies response.tooDark = false \cap response.inFrame = true \cap response.kneeVisible = true) \cup response.ok = false \implies response.message \text{ describes failure reason}$

resetBackend():

Pre-condition: $pose_app \neq \emptyset$

Function: *resetBackend*()

Post-condition: $pose_app.analyzer = newAnalyzer() \cap c = true \cap cal_time_left = 10s \cap rep_count = 0 \cap min_degree, max_degree, rom_degree = 0$

server.py

reset_backend (POST/reset endpoint)

Pre-condition: $pose_app \neq \emptyset$

Function: *reset_backend*()

Post-condition: $pose_app.analyzer = newAnalyzer()$

$is_calibrating = true$

$cal_time_left = 10.0$

$rep_count = 0$

precheck_frame (POST/precheck_frame endpoint)

Pre-condition: $\forall j \in J : j.imageBase64 \in b \cap j.side \in s \cap j.mirrored \in true, false \cap cameraReady = true$

Function: *precheckFrame*()

Post-condition: $response.message \neq \emptyset : (response.ok = true \implies response.tooDark = false \cap response.inFrame = true \cap response.kneeVisible = true) \cup response.ok = false \implies response.message \text{ describes failure reason}$

process_frame (POST/process_frame endpoint)

Pre-condition: $\forall j \in J : j.imageBase64 \in b \cap j.side \in s \cap j.mirrored \in \{true, false\} \cap cameraReady = true \cap pose_app.analyzer \neq \emptyset$

Function: *processFrame*()

Post-condition: $\{rep_count\}(t_2) \geq \{rep_count\}(t_1) \forall t_2 > t_1, rom_degree = max_degree - min_degree \geq 0 \cap angle \in [0, 180] \cap c(t_2) = false \implies c(t_1) = false \cup cal_time_left(t_1) \leq 0$

6.7 Module - Analyze

6.7.1 Uses

analyze.py - Variables

`min_angle` Stores the minimum knee angle observed during calibration of range of motion.

`max_angle` Stores the maximum knee angle observed during calibration of range of motion.

`rep_count` Tracks the number of completed repetitions.

`rep_state` Current phase of repetition ("Extension", "Flexion").

`cue_state` Current feedback cue state ("CALIBRATING", "GOOD_EXTENSION", "GOOD_FLEXION", "GETTING_READY", "OUT_OF_FRAME").

`smoothed_angle`
Exponentially smoothed angle value to reduce jitter.

`cycle_min_angle`
Minimum angle recorded within the current repetition cycle.

`cycle_max_angle`
Maximum angle recorded within the current repetition cycle.

`last_rep_time`
Timestamp of the last completed repetition.

analyze.py - Types

None

analyze.py - Functions

`_smooth_angle(angle)`
Applies smoothing to angle measurements/calculates acceptable angle values.

`_calibrate_with_angle(angle)`
Updates min/max angle values during calibration window.

`_rep_update(angle)`
Core repetition detection logic using zone-based thresholds.

`start_calibration(duration_s)`
Begins calibration phase for specified duration in seconds.

`update(angle)` Main entry point for processing each new angle measurement.

`calc_rom()` Returns range of motion as `max_angle` minus `min_angle`.

`summary()` Returns dictionary containing all current analysis metrics.

server.py - Variables

`pose_app` Global singleton managing analyzer state and camera.

`analyzer` Instance of Analyzer class accessed via `pose_app.analyzer`.

`lock` Ensures thread-safe access to analyzer during concurrent requests.

server.py - Types

None

server.py - Functions

`Flask` Creates the Flask web application instance and defines API routes.

`jsonify` Converts Python dictionaries to JSON HTTP responses.

`request` Accesses incoming request data including JSON body.

`b64_to_bgr_image(b64_str)`
Decodes base64 string to OpenCV BGR image format.

`process_frame()`
Receives frame, calculates knee angle, updates analyzer, returns metrics.

`check_side_visibility(landmarks, side)`
Determines if specified knee and adjacent joints are visible in frame.

`Thread` Enables concurrent request handling via Flask's `threaded=True`.

6.7.2 Syntax

Exported Constants

analyze.py, server.py

None

Exported Access Programs

analyze.py

Input	Output	Exceptions
start_calibration(duration_s): Begins calibration phase to record min and max knee angles.		
duration_s: float	None	None
update(angle): Main entry point for processing each new angle measurement. Updates smoothed angle, calibration state, and rep detection.		
angle: float	None	None
calc_rom(): Calculates and returns the current range of motion.		
None	float - Range of motion in degrees	None
summary(): Returns a dictionary containing all current analysis metrics.		
None	dict - Contains min_degree, max_degree, rom_degree, calibrating, cal_time_left, rep_count, current_rep_duration, avg_rep_duration, rep_state, cue_state	None

Table 10: Access Routines for Analysis Module

server.py

None

6.7.3 Semantics

State Variables

S: set containing "Extension" and "Flexion"

ex_state: Exercise state

C: "CALIBRATING", "GOOD_EXTENSION", "GOOD_FLEXION", "GETTING_READY", "OUT_OF_FRAME"

started: (True, False) indicating whether exercise has started.

angle: angle measurement of landmark vectors

min_angle: min_angle

max_angle: max_angle

is_calibrating: (True, False) indicating whether calibration is taking place.

cue: exercise feedback string

A: set of Analyzer instances

State Invariants

$\forall angle \in \{angles\}, angle \in 0 \leq angle \leq 180$

$$\begin{aligned}
& (\text{started} = \text{True} \implies \text{is_calibrating} = \text{False} \cup (\text{ex_state} \in S \mid \text{ex_state} = \text{"Flexion"} \cup \\
& \text{"Extension"})) \cup (\text{started} = \text{False} \implies (\text{is_calibrating} = \text{True})) \\
& \forall \text{ex_state} \in S, \text{cue} \in C, \text{ex_state} = \text{"Extension"} \implies \text{cue} = \text{"GOODEXTENSION"} \cup \\
& \text{ex_state} = \text{"Flexion"} \implies \text{cue} = \text{"GOODFLEXION"} \\
& \forall a \in A, |a| = 1
\end{aligned}$$

Environment Variables

NA

Assumptions

- Basic process threading rules apply when using an Analyzer instance. Only one process can function properly.

6.8 Module - Feedback

6.9 Module - Metrics

6.10 Module - Draw

6.11 Module - UI

References

Carlo Ghezzi, Mehdi Jazayeri, and Dino Mandrioli. *Fundamentals of Software Engineering*. Prentice Hall, Upper Saddle River, NJ, USA, 2nd edition, 2003.

Daniel M. Hoffman and Paul A. Strooper. *Software Design, Automated Testing, and Maintenance: A Practical Approach*. International Thomson Computer Press, New York, NY, USA, 1995. URL <http://citeseer.ist.psu.edu/428727.html>.

7 Appendix A

Table 11: Database Schema

Tables	Fields
activities	actid, createdAt, duration, email, exercise, timestamp, uid, analysis, avg_rep_duration, completed_reps, completed_sets, current_angle, current_rep_duration, date_performed, effort, max_height, min_height, name, pain, patient_feedback, rep_state, rom_degree, satisfaction, target_area, Database location
comments	actid, author, cid, comment, date, uid
exercises	description, reps, sets, title
inviteCodes	active, physioId, used, usedBy
userExercises	description, id, reps, sets, title, userid
users	birthday, createdAt, email, inviteCode, name, physioId, role
validLicenses	active

Appendix — Reflection

The information in this section will be used to evaluate the team members on the graduate attribute of Problem Analysis and Design.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which of your design decisions stemmed from speaking to your client(s) or a proxy (e.g. your peers, stakeholders, potential users)? For those that were not, why, and where did they come from?
4. While creating the design doc, what parts of your other documents (e.g. requirements, hazard analysis, etc), if any, needed to be changed, and why?
5. What are the limitations of your solution? Put another way, given unlimited resources, what could you do to make the project better? (LO_ProbSolutions)
6. Give a brief overview of other design solutions you considered. What are the benefits and tradeoffs of those other designs compared with the chosen design? From all the potential options, why did you select the documented design? (LO_Explores)
7. What have you learned from implementing another team's module?

Group - Reflection

1. N/A
2. N/A
3. N/A
4. I think that our other documents such as the [requirements](#), [hazard analysis](#) need to be slightly adjusted as a result of the design documents. With the requirements specifically some of them pertain to how the application design is and after completing our initial draft of the MG and MIS, the design does not necessarily line up with the requirements document. Further, some of the hazards considered should be changed to include a couple more areas for hazards to occur. Lastly for both of the documents, the assumptions should be changed to incorporated design changes that we had not considered during that deliverable.
5. Some limitations on the current system design and with the set scope include the accuracy of the pose estimation, the responsiveness of the application, and the limitation of stored exercises. With more resources, we would like to improve the accuracy of the pose estimation by either using a more advanced model or training our own model specifically for physiotherapy exercises. Further, we would like to improve the responsiveness of the application by optimizing our code and potentially using more powerful hardware to run the application. And finally, although explicitly stated in our scope, we would like to include more exercises and potentially allow for custom exercises to be added by physiotherapists.

6. One alternative design solution we considered was making the application a desktop application rather than an android application. The benefit of this design is that it would allow for more powerful hardware to be used, which could improve the responsiveness and accuracy of the application. However, the tradeoff is a limitation to the accessibility of the application, as not all users may have access to a desktop computer with hardware capable of recording. We ultimately chose the android application design because it allows for greater accessibility and convenience for users, as they can use the application on their mobile devices which are easier to setup and come with recording hardware.
7. Implementing another team's module made it clear that another level of description is needed in order to effectively communicate the design and function of each module to developers. While implementing the Pixelation module for team 7, we found the description of their access program inputs and outputs lacking in detail. With sufficient context from the rest of the documentation we were able to produce what we believed to be the intended functionality of the module, but it required a lot of extra time and effort to interpret the design document. Additionally, because of this vague input and output description, there were no names given to input and output variables, which made it harder to stick to their coding conventions but also even harder to interpret the context of the module.

There was a lack of detail which I feel resonates with our documentation as well. The purposes and usage of each module by other modules should be clearly defined in order to avoid confusion during implementation. Working on the final version of the documentation, we will make sure to include detailed descriptions of each access program's inputs and outputs, as well as any necessary context for understanding the module's function within the overall system.

There was an overall description of modules and their functionality which provided a good high-level overview of each module, something we think should be included in our documentation as well. Additionally the use of assumptions to clarify certain design decisions was helpful, and while we don't believe it to be the correct section to provide context it was a good signifier of a well made assumption.

Overall, this experience has taught us the importance of clear and detailed documentation in order to facilitate effective communication and collaboration between teams.

Matthew - Reflection

1. I think something that went well was our changes to overall workflow. One thing that we recognized was starting our document on google docs and then transferring our sections to GitHub on the last two days the the deliverable was due led to some things going unchecked and the deliverable not turning out the way we had planned. Further it had also resulted in a 'panic' when we had to resolve merge conflicts between our sections soon to the deadline. Overall, we cut google docs for this deliverable and

mainly focused on our sections within GitHub and doing multiple small PRs within the deliverable timeline and that had gone quite well. There are a few days left but a majority of the document has at least been accounted for and hopefully this workflow works for our team.

2. I think one pain point I experienced was time management. I had suggested that since the computing team's workload was quite different to the documentation team's, we could take on small sections within the document with hopes of submitting it early into the project. This did partially go well but with focus split in two areas it was hard to focus on the development side of things for the POC demo. This may be something that improves with time but as of now is currently a work in progress as development on my end is not where I want it to be.
3. I think that our design decisions partially came from speaking with our stakeholder (Kevin Yu) but other decisions were made with typical application creation. For example, the high-level was talked through with Kevin to simulate the timing and feedback delivery that you would encounter during physiotherapy but things like reporting, logging-in as well as account creation were from looking at other applications and aspects that went well from that end.
4. Group Q
5. Group Q
6. Group Q

Vaisnavi - Reflection

1. Something that went well was how we worked on creating more constant pull requests (PRs) to avoid merge conflicts. This worked out well with everyone consistently pushing more PRs early on, allowing someone on the team to review and effectively merge it in. Overall, the organization and collaboration went very smoothly, and it allowed us to ultimately finish ahead of our internal deadline so we had enough time for editing and formatting.
2. I think one of the major pain points through this deliverable was trying to balance both the development side for the POC demo while also trying to contribute to the document. As the main sub-team for the POC demo, we did take on smaller sections, but it was difficult to find that balance in time for both at the start.
3. We believe that most of our major design decisions stemmed directly from our meetings with our stakeholder, Kevin Yu. Throughout the development of this deliverable, a number of ideas that we originally planned had to be changed or completely re-worked based on the feedback we received. In several cases, Kevin helped us recognize that certain features were either not feasible within the course timeline or did not align with

the true purpose and scope of our project. This was extremely helpful overall to have this input to tackle these decisions early on in the process.

4. Group Q
5. Group Q
6. Group Q

Maham - Reflection

1. Despite our (extremely) busy schedules, we made time to meet when we could and offer assistance to any member that needed it. In addition, all of us kept in contact and checked in with each other. I liked that none of us completely abandoned this project, but rather balanced it quite well. Also, the more frequent PR requests strategy (implemented for this deliverable) ended up being immensely helpful.
2. A pain point for me was that a lot of my time and energy was taken up by external tasks. These were, unfortunately, appointments I could not compromise on because they had been scheduled months in advance. I would have loved to give this project more of my focus, and I tried my best to self-organize, but there were some areas I fell short. I will attempt to refine my balancing act for the future.
3. The project is still in the early stages of development, and as a result, we do not have many concrete elements. However, some of developed ideas and design decisions was aided by Kevin, our supervisor. At this stage, we are constructing the foundation of the software, and that knowledge was supplied by him. Eventually, when we do have a functional product, we will want to loop in other stakeholders, e.g. patients, for modifications
4. Group Q
5. Group Q
6. Group Q

Cieran - Reflection

1. During this deliverable there was a need to agree upon implementation tactics. The team met and nailed down an approach for the development of the proof of concept, and I think everyone did a good job of ensuring their thoughts on what the system *was* got at least mentioned when developing the system.
2. I yet again sharked my teammates with procrastination. I have apologized profusely for my lack of initiative with this deliverable and have taken steps to ensure progress is far more consistent and meaningful in the future. As for this deliverable, I worked to catch up my own lack of effort towards the end of the it.

3. I think the design decisions, specifically around the system's allowance for physiotherapists to alter patient's exercise routines, were very influenced by meetings with our advisor. As a team, we wanted to ensure that the system was something wanted by both patients and physiotherapists; designing based off of requirements from past physiotherapy patients and our physiotherapist advisor was a must. Some design decisions, such as checking against public databases for the physiotherapist's identification number, were made in an attempt at security for users of the application.
4. GROUP Q
5. GROUP Q
6. GROUP Q